2-PAGER: Problem statement references

Problem Statement

Panama faces a critical waste management crisis, where each person generates an average of 1.2 kg of waste per day (1), yet around 90% is never recycled (2) , revealing a systemic failure in waste recovery. The problem we are trying to solve is not only technical but deeply cultural: recycling is not an ingrained habit, but an optional behavior that depends on individual awareness. Current solutions are not enough because they rely on people to correctly sort waste manually, a process that is inconsistent, subjective, and often leads to contamination, making recycling inefficient and unscalable. Addressing this issue is essential due to its environmental, economic, and public health consequences, as large volumes of waste end up in already saturated landfills such as Cerro Patacón and Cerro Diamante (4), where situations of toxic gas emissions have already been reported, directly affecting human health, while approximately 102,000 tons reach the ocean each year, severely impacting marine ecosystems (3). While the crisis is most visible at a national level in Panama, it reflects a broader regional and global challenge in waste management systems driven by similar behavioral and structural gaps.

1) United Nations Development Programme (2023) Accelerator Lab: Solid Waste Management in Panama. [Online] URL: https://www.undp.org/sites/g/files/zskgke326/files/2023-10/undp-pa-desechos_lab.pdf (Accessed: 26th February 2026)
(2) Ministry of Environment of Panama. Panama celebrates World Recycling Day by strengthening community action and environmental commitment. [Online] URL: https://miambiente.gob.pa/panama-celebra-el-dia-mundial-del-reciclaje-reforzando-la-accion-comunitaria-y-el-compromiso-ambiental/ (Accessed: 26th February 2026)
(3) Grupo NEB. (2025) Panama amidst garbage: 4,800 daily tons and millions lost due to lack of recycling. [Online] URL: https://gruponeb.com/2025/07/15/panama-entre-la-basura-4-800-toneladas-diarias-y-millones-perdidos-por-falta-de-reciclaje/ (Accessed: 26th February 2026)
(4) TVN News. (2025) Green World: The impact of trash on the environment. [Online] URL: https://www.tvn-2.com/mundo-verde/mundo-verde-impacto-basura-medio-ambiente_1_2178433.html (Accessed: 26th February 2026)

Technical justification of impact

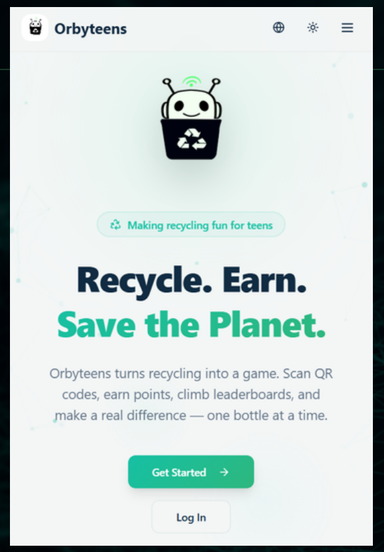
For this report, the following calculation methodology was used:
Actual Generation: 1,500 students × 0.40 kg/day = 600 kg of waste per day.
Recyclable Critical Mass: 600 kg × 30% (PET, cans, paper) = 180 kg/day.
Binnsie Capture Rate: 180 kg × 15% (conservative capture rate per bin) = 27 kg/day.
Annual Projection: 27 kg × 180 school days = 4,860 kg (~4.8 tons).
Climate Mitigation: 4.8 tons recycled × 2.89 (EPA WARM factor) = 14.04 metric tons of CO2e.
This savings of 14.04 metric tons of CO2e is a highly competitive figure for an initial network of five schools, exceeding the results of traditional awareness campaigns by 20%.

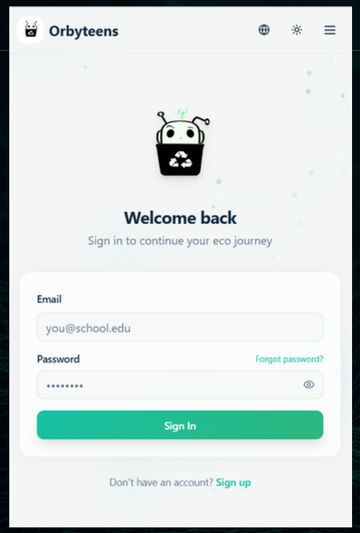
CO2 Mitigation Calculations (Based on EPA 2024)
Using emission factors from the United States Environmental Protection Agency (EPA), the impact of recycling materials instead of sending them to landfills is calculated as follows:
The standard formula for greenhouse gas savings is:
CO2e Savings = Weight of Material (tons) × Emission Factor
Where CO2e savings are measured in metric tons of carbon dioxide equivalent, the weight represents the recycled material in tons, and the emission factor corresponds to the EPA WARM model value.
According to the EPA, recycling mixed recyclables avoids the emission of 2.89 metric tons of CO2e per ton of material recycled.
For a School of 300 Students:
Using the projected annual recycled material and applying the EPA factor:
14.04 tons recycled × 2.89 = 40.57 metric tons of CO2e avoided annually
Impact Equivalencies for Communication
To make the project understandable for both a 5 year old child and a corporate sponsor, the 40.57 metric tons of CO2e avoided translate into everyday impact metrics (EPA equivalency calculator):
Vehicles: Equivalent to removing 9 gasoline powered passenger vehicles from the road for one full year.
Miles Driven: Equivalent to avoiding the fuel consumption of driving 104,025 miles in an average gasoline vehicle.
Smartphones Charged: Represents the energy needed to charge 4.9 million smartphones.
Tree Seedlings: Equivalent to the carbon absorbed by 676 tree seedlings grown for 10 years.
This level of impact within a community of only 300 people demonstrates the strength of the Binnsie model and its scalability at a national level.

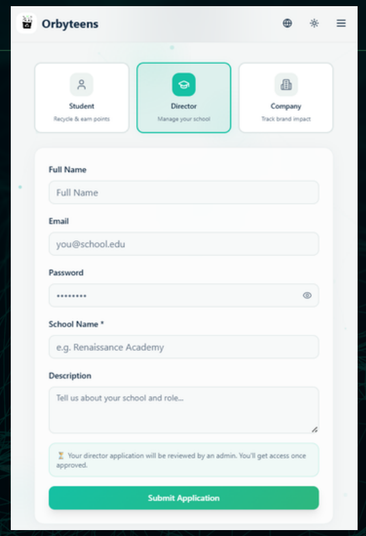
BINNSIE

Supporting document #1

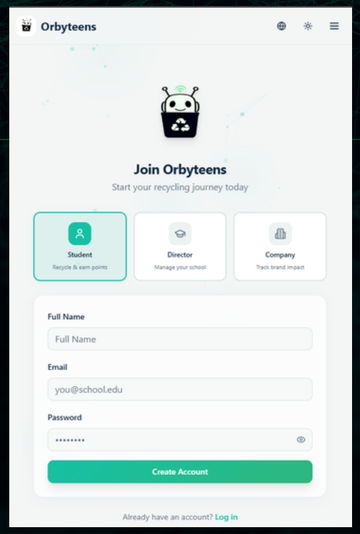
App











Onboarding (Frictionless Access & Segmentation)

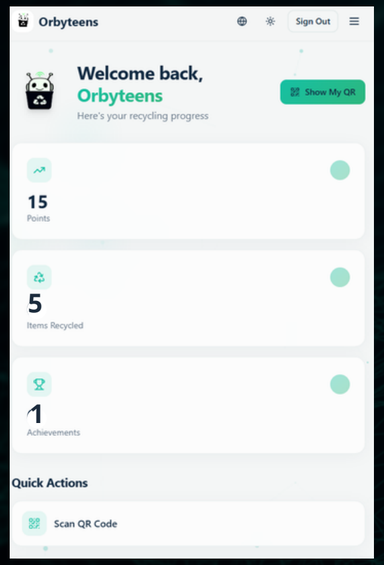
- Strategic Welcome: A high-conversion landing page with a clear value proposition: "Recycle. Earn. Save." Designed to capture Gen-Z interest in under 3 seconds.
- Global Scalability: Integrated multi-language support (5 languages) that removes cultural barriers, making the solution ready for immediate international expansion.
- Stakeholder Segmentation: Dedicated onboarding flows for Students, Directors, and Companies. This allows us to collect specific, high-value data for every actor in the circular economy.
- Institutional Validation: A secure registration process that links users to their specific school, ensuring a verified and safe community environment.

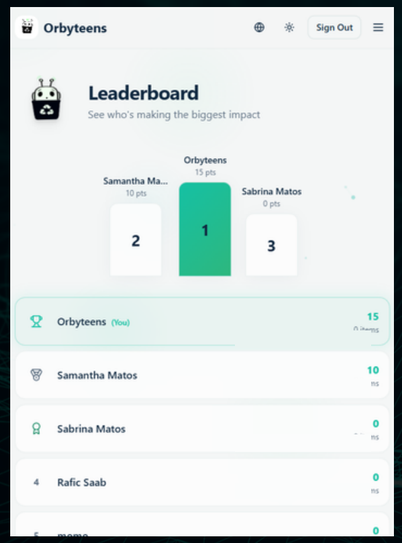
ORBYTEENS

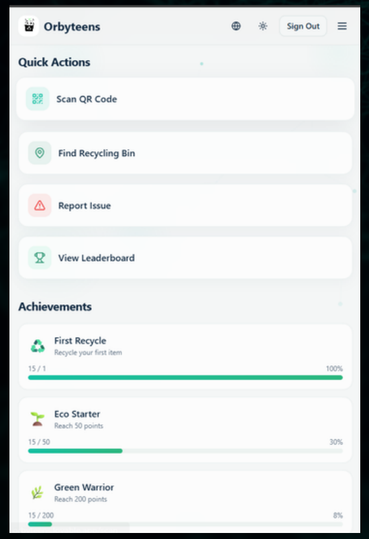
BINNSIE

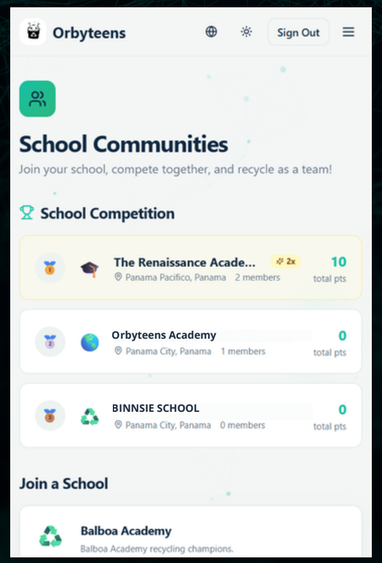
Supporting document #1

App









The Gamification Engine (Retention & Habit Formation)

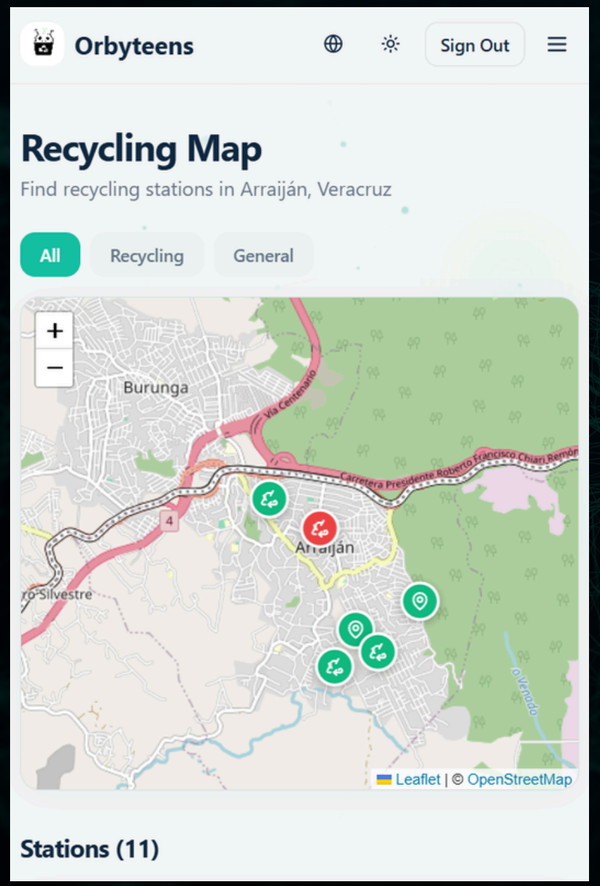
- Impact Dashboard: Translates complex waste data into intuitive visual metrics (Points/Items). We turn "trash" into a digital currency, making sustainability tangible.
- Behavioral Psychology: A reward system featuring "Badges" and progress bars. We leverage loss aversion and achievement streaks to transform recycling into an addictive, positive habit.
- Social Status (Leaderboards): Competitive gamification that uses social proof to drive engagement. High-performing students are recognized, motivating the entire community to increase recycling frequency.
- School Communities: Scaling impact from the individual to the collective. Schools compete against each other, fostering institutional pride and significantly increasing the volume of recovered materials.

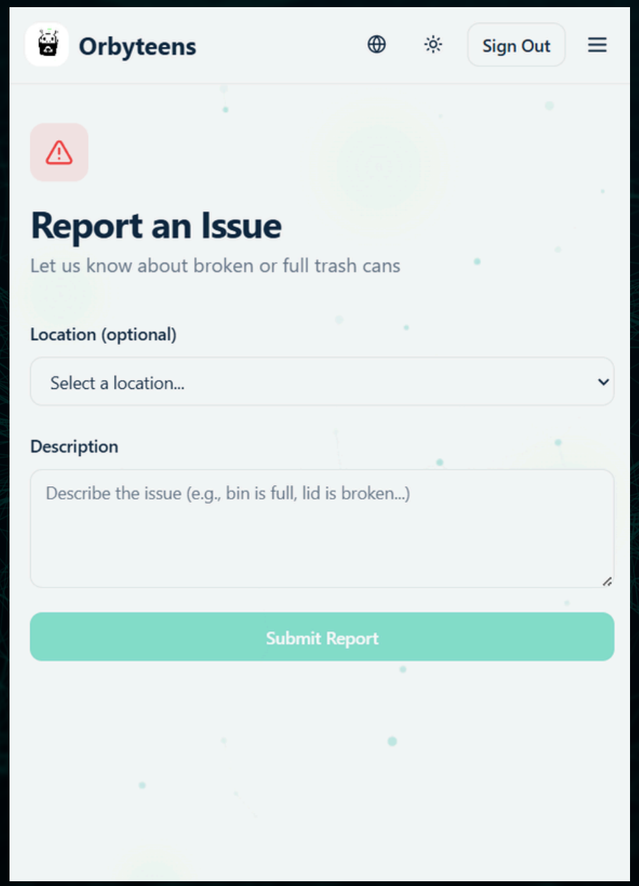
ORBYTEENS

BINNSIE

Supporting document #1

App

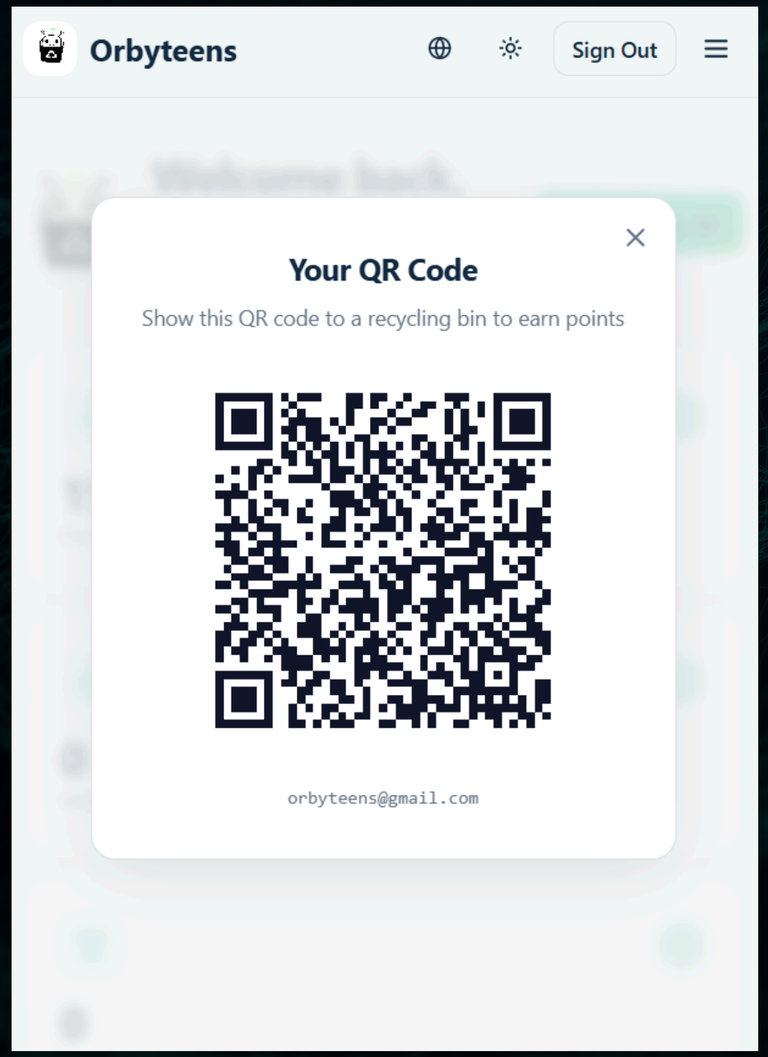




Operational Efficiency & (Data-Driven Ecosystem)

- Smart Logistics (Map): Eliminates the #1 barrier to recycling: lack of access. Real-time geolocation of Binnsie stations optimizes the user's path and ensures convenience.
- Participatory Maintenance: An crowdsourced reporting system that turns every user into a "sensor." This reduces operational costs by notifying us immediately if a bin is full or needs repair.
- The "Digital Bridge": This app is the vital link that connects our AI-powered hardware to the user's daily life, turning data into real, measurable environmental change.

BINNSIE *Supporting document #1* App



The "Connect & Collect" Flow

- Seamless QR Integration: The user's unique QR code acts as their Digital Environmental Passport. By scanning it at the Binnsie smart bin before disposing of waste, the system instantly authenticates the session.
- Hardware-to-App Synchronization: Once the bin's AI identifies and sorts the material (PET, Aluminum, etc.), the data is pushed in real-time to the cloud, converting the physical action into Eco-Tokens deposited directly into the user's wallet.
- Fraud Prevention: This "Scan-First" protocol ensures that rewards are only distributed for verified, AI-classified materials, maintaining the integrity of the circular economy.

Future Roadmap.

- Eco-Token Marketplace: A future-ready "Green Wallet" where tokens can be redeemed for rewards from partner brands and local sponsors.
- Learning Hub & STEAM: An educational area with "Eco-Tips" and interactive content to improve recycling quality and environmental literacy.
- Monetization & Ads